← Back **Systems of Differential Equations Graded Assignment**
Graded Assignment • 8 min

Due Nov 11, 11:59 PM IST

Your grade: 100%

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1. **True** or **False**: Suppose that *A* is an $n \times n$ matrix and $\mathbf{y} = e^{\lambda t}\mathbf{u}$ is a solution to the system of linear differential equations $\mathbf{y}' = A\mathbf{y}$ where $\mathbf{u}$ is an eigenvector of *A* with associated eigenvalue λ. Let *k* be any scalar. Then $\mathbf{y} = ke^{\lambda t}\mathbf{u}$ is a solution to the system $\mathbf{y}' = kA\mathbf{y}$. 1 / 1 point

   - True
   - False

   Nice work

2. **True** or **False**: Suppose that *A* is an $n \times n$ matrix and $\mathbf{y} = e^{\lambda t}\mathbf{u}$ is a solution to the system of linear differential equations $\mathbf{y}' = A\mathbf{y}$ where $\mathbf{u}$ is an eigenvector of *A* with associated eigenvalue λ. If *A* is invertible, then $\mathbf{y} = e^{t/\lambda}\mathbf{u}$ is a solution to the system $\mathbf{y}' = A^{-1}\mathbf{y}$. 1 / 1 point

   - True
   - False

   Nice work

3. **True** or **False**: If $a, b, r, q > 0$, and y_1, y_2 are solutions to the initial-value problem $\begin{matrix} y_1' = -ay_1 + by_2 \\ y_2' = ay_1 - by_2 \end{matrix}$, with $y_1(0) = r,$ and $y_2(0) = q$, then $y_1(t) + y_2(t) = r + q$ for all *t*. 1 / 1 point

   - True
   - False

   Nice work

4. **True** or **False**: If *A* is a real square matrix with complex eigenvalue λ and associated eigenvector $\mathbf{u}$, then $\mathrm{Im}(e^{\lambda t}\mathbf{u})$ is a real solution to the system $\mathbf{y}' = A\mathbf{y}$. 1 / 1 point

   - True
   - False

   Nice work